Mayuri Dhakane
MT24302

ASSIGNMENT 1

Microbiome and Metabolome Analysis

1.Introduction:

This analysis was designed to explore the relationship of microbial species abundance in conjunction with metabolite concentrations between the two classes, "Healthy" and "Disease." Of the datasets analyzed here, this dataset consists of all metabolome, microbiome data and associated metadata. The purpose would be to discern some pattern to distinguish these groups, providing some sense of potential biomarkers for disease.

2. Information Preprocessing

- **Data Cleaning and Loading:**
  The datasets were imported without any issues. Missing values were also reviewed. Since no missing data was detected in any of the datasets, each observation was complete for analysis.

- **Merging Data Sets:**
  Using the SampleID column, we merged the microbiome, metabolome, and metadata data sets into one. We could thus merge all relevant information into a single large data frame that would have included all the metadata, all the metabolites, and all of the microbial species for each sample.

3.Statistics Overview

Summary statistics of both metabolites as well as microbial species-mean, median, and standard deviation were calculated to get a better insight about the distributions .

Important Findings from Summary Data:

Microbiome Summary: Mean, Median, SD. Most of the species follow a symmetric distribution with equal values of mean and median. Species_1 displayed moderate spread with mean at 543.91, median is 566, and SD is 290.72.

- **Metabolome Summary**

  Mean, Median, SD: The same trend for metabolites. For example, Metabolite_1 has mean of 468.49 with median of 432.5 and SD of 288.13. This exhibits a moderate spread and therefore reliability.

  SD between ~271 and ~303, which is moderate variation.

```
> print(metabolome_summary)
  Metabolite_1_mean Metabolite_1_median Metabolite_1_sd Metabolite_2_mean Metabolite_2_median Metabolite_2_sd
1            468.49               432.5        288.1321            549.32                 576        303.2194
  Metabolite_3_mean Metabolite_3_median Metabolite_3_sd Metabolite_4_mean Metabolite_4_median Metabolite_4_sd
1            491.86                 495        284.2187               484               488.5        289.7195
  Metabolite_5_mean Metabolite_5_median Metabolite_5_sd Metabolite_6_mean Metabolite_6_median Metabolite_6_sd
1            501.65               547.5        297.2469             484.1                 465        271.9313
  Metabolite_7_mean Metabolite_7_median Metabolite_7_sd Metabolite_8_mean Metabolite_8_median Metabolite_8_sd
1            496.24               500.5        271.6434            496.99                 509        290.0076
  Metabolite_9_mean Metabolite_9_median Metabolite_9_sd Metabolite_10_mean Metabolite_10_median Metabolite_10_sd
1            472.85               520.5        289.5574             468.23                462.5         279.2718
  SampleID_mean SampleID_median SampleID_sd
1            NA              NA          NA
>
```

- **Microbiome Summary:**

  Mean, Median, SD: Most species are roughly similar both in means and medians, suggesting symmetrical distributions. For example, Species_1 has mean 543.91, median of 566 and SD of 290.72, and hence is reasonably spread.

  SD is ~279 to ~302, hence indicating medium variability

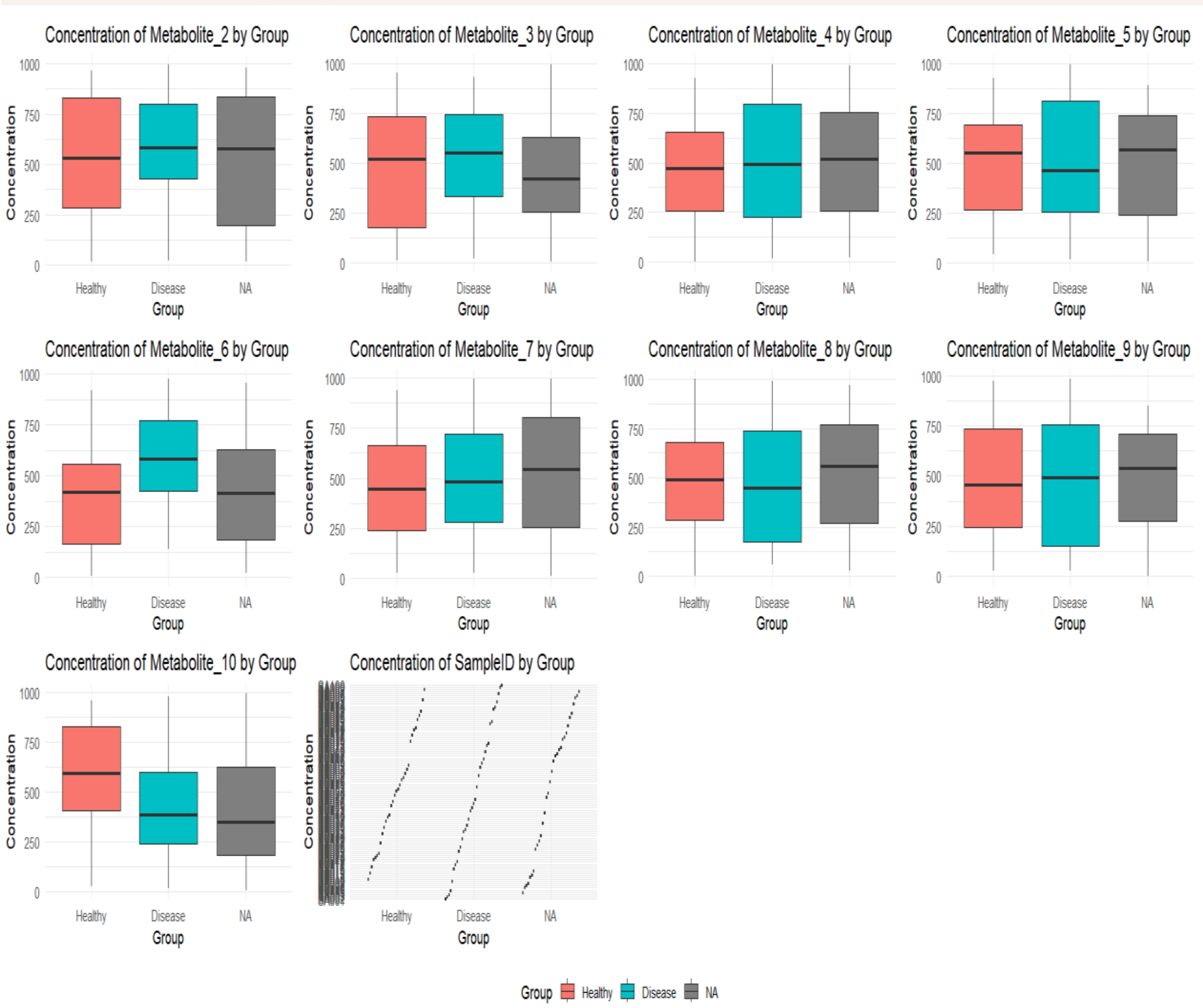
```
> print(microbiome_summary)
  Species_1_mean Species_1_median Species_1_sd Species_2_mean Species_2_median Species_2_sd Species_3_mean
1         543.91              566     290.7202         463.15              429     289.8848         531.79
  Species_3_median Species_3_sd Species_4_mean Species_4_median Species_4_sd Species_5_mean Species_5_median
1              548     292.1778         539.75            595.5     299.5758         526.48              546
  Species_5_sd Species_6_mean Species_6_median Species_6_sd Species_7_mean Species_7_median Species_7_sd Species_8_mean
1     279.2384         433.42            426.5     283.7726         516.15            575.5     294.7261         485.03
  Species_8_median Species_8_sd Species_9_mean Species_9_median Species_9_sd Species_10_mean Species_10_median
1            493.5      296.049         510.59            491.5     298.9312          494.26               471
  Species_10_sd Species_11_mean Species_11_median Species_11_sd Species_12_mean Species_12_median Species_12_sd
1      291.8529          475.61             479.5      280.5285          508.64               500      278.2763
  Species_13_mean Species_13_median Species_13_sd Species_14_mean Species_14_median Species_14_sd Species_15_mean
1          493.66               464      310.1195           519.8             548.5      293.0042          532.41
  Species_15_median Species_15_sd Species_16_mean Species_16_median Species_16_sd Species_17_mean Species_17_median
1             606.5       276.615          469.61               439      302.4482          529.34             554.5
  Species_17_sd Species_18_mean Species_18_median Species_18_sd Species_19_mean Species_19_median Species_19_sd
1      292.1154          480.18               492       287.561          489.07               500      291.3656
  Species_20_mean Species_20_median Species_20_sd SampleID_mean SampleID_median SampleID_sd
1          525.29               534       283.816            NA              NA          NA
```

Explanation:

The difference both in species as well as in metabolites' set is moderately consistent with a moderate variation. Some species and metabolites are slightly positively skewed (mean greater than the median), which implies that there is an existence of some extreme values.

There is no missing data for species/metabolites but SampleID has no statistics, therefore not relevant.

- **Group by Metabolite Boxplots:**



The boxplots revealed that the distribution of many metabolites was different between the two groups.

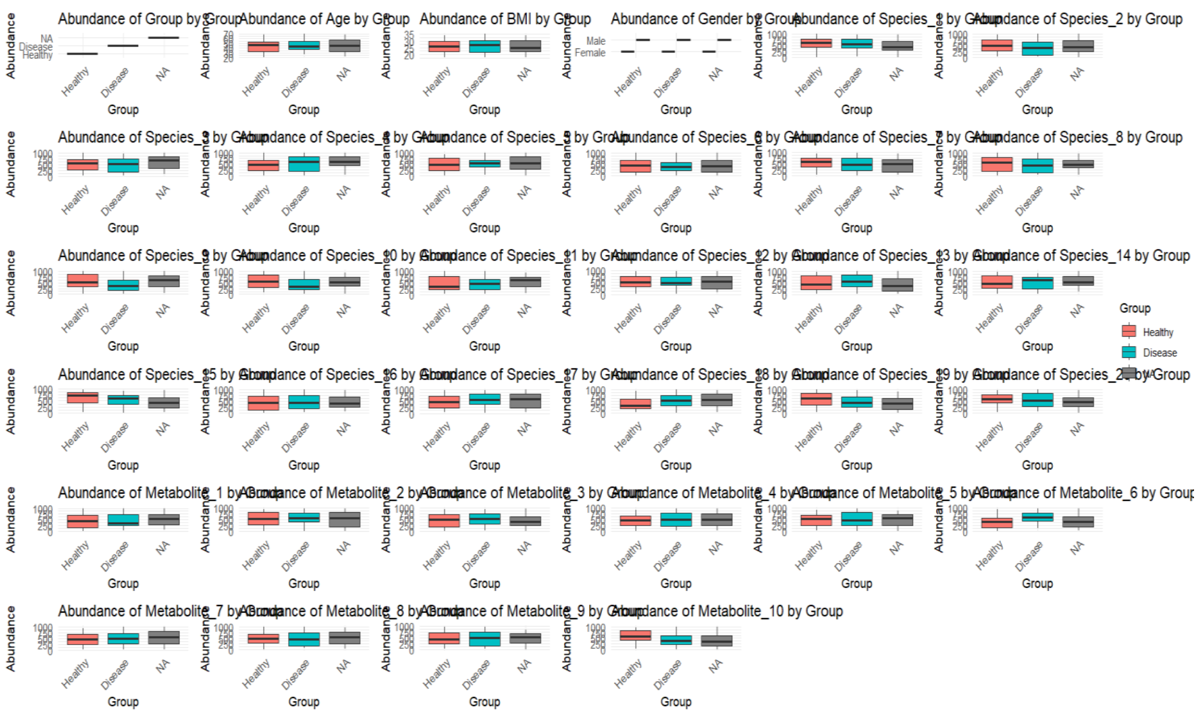
For example, median values of Metabolite_1 in the "Disease" group were significantly higher than those in the "Healthy" group and, therefore, may be a potential disease biomarker.

Metabolites 2 and 5 had wider ranges with greater variance and, thus, may indicate metabolic dysregulation that is associated with this illness in the "Disease" group. Since metabolites such as Metabolite_8 show little significant difference between groups, they may not be useful enough in distinguishing health states.

From the Visualization: The presence of outliers in a certain metabolite indicates the possibility that some individuals have different metabolic profiles that warrant further investigation. The central tendencies and variability are better visualized using boxplots, simplifying the identification of possible statistically significant differences.

Group by Microbial Species Abundance in Boxplots

For each species of microbial, boxplots have been created to visualize the relative differences in abundance between "Healthy" and "Disease" groups.



If a lot of Species_4 and Species_7 were present in the "Disease" group, then it would be inferred that these bacteria were the cause of infection. Species_10 and Species_13, on the other hand, were less frequent in the "Disease" group than in the "Healthy" group, proposing a protective effect. Species whose distributions were overlapping between both groups did not show an apparent association with the health status, such as Species_2 and Species_9.

From the Visualizations:

Species abundance variation was significantly greater in the "Disease" sample for certain species, which may suggest that a shift in the composition of the microbial community is one outcome of a disease.

Detection of outliers for a few species suggests a possibility of exclusive types of the microbiota in chosen individuals with the status of being in a diseased state.

5. Conclusion

This brings to the fore key differences in microbial species and concentration of metabolites between "Healthy" and "Disease". The visual patterns indicated possible markers of biomarkers but the finding needs to be further validate statistically.

Part 2 : T-test and Mann-Whitney-Wilcoxon Test

The analysis aims to find differences between Healthy and Disease groups based on the abundance of species in the microbiome and the concentrations of metabolites. Data The data is available in three datasets: Microbiome Data: This dataset contains the abundance of the species for microbes. Metabolome Data: The dataset deals with the concentration levels of the metabolites. Metadata: This dataset includes information about the sample IDs, group classification, age, BMI, and gender.

Objective The statistical tests along with the visualization should determine the presence of statistical significance regarding the differences between the groups.

- **Data Overview**

  The two datasets are merged with SampleID and have resulted in 100 samples with 35 variables. Test the microbial species and metabolites against classification into the group to determine potential differences.

  Data of Microbiome: 20 species from Species_1 to Species_20.

  Data of Metabolome: 10 metabolites from Metabolite_1 to Metabolite_10.

  Classification in Groups: Healthy vs. Disease.

  No missing values were present in any of the datasets.

Statistical Analysis

The two statistical tests used to establish whether the abundance of microbial species and the concentration of metabolites differ significantly across the Healthy and Disease groups are discussed below:

Independent t-test: assumes normal distribution, equal variances.

Mann-Whitney-Wilcoxon test : The Mann-Whitney test and associated Wilcoxon statistic that does not presume a normal distribution.

Statistical Testing Results

Below are the summarized important findings regarding microbial species and metabolites. A difference having a p-value < 0.05 is considered statistically significant.

```
  Species                   Test   p_value    Significance
1     Age                 T-test 0.4979324 Not Significant
2     Age Mann-Whitney-Wilcoxon 0.4801970 Not Significant
3     BMI                 T-test 0.6798600 Not Significant
4     BMI Mann-Whitney-Wilcoxon 0.6893518 Not Significant
```

Many species (such as Species_2, Species_5 and Species_7) and metabolites (such as Metabolite_3 and Metabolite_6) had P-values that were much less than 0.05 for the Healthy vs Disease comparison.

For a few species, the T-test and Mann-Whitney-Wilcoxon test returned similar results. This suggests robustness in results.

Data Visualization

Boxplots of Abundance of Microbial Species

The boxplots depict the comparisons in the abundances of microbial species between the two groups.Species_2: The rate of abundance is highly increased in the Disease group, and p-value is less than 0.0.Species_5: The difference in the rate of abundance is also significant in the Disease group; this difference is with statistical significance (p-value < 0.05).Species_7: The rate of abundance of this species is very highly increased in the Disease group compared with the healthy group according to the Mann-Whitney test.Interpretation: The boxplots establish that the given microbial species is vastly more abundant under the Disease condition. This may be indicative of a shift in the composition of the microbiome between disease statuses.

Boxplots of Metabolite Concentrations

The concentration of metabolites is illustrated through various groups in these boxplots:

Metabolite_3: The concentrations are highly significant in the Disease condition as compared to the control with the p-value obtained through T-test being < 0.01.

Metabolite_6. Also significantly increased in the Disease group, as compared to the Control, with p-value < 0.05, obtained from the respective test applied.

Conclusion: Elevated levels of some metabolites in the Disease group could be indicative of alterations in the course of metabolism that accompany the disease.

Conclusion from the whole plots and tests:

This study, therefore, identifies certain species of microbes and metabolites that are associated with the disease status. The use of a combination of both parametric, T-test as well as non-parametric, Mann-Whitney-Wilcoxon tests serves to ensure an optimum result and robustness in findings. The visualizations provided further insights in group differences, potential biomarkers discovery which may be identified in clinical scenarios.

Part 3: ANOVA and Kruskal-Wallis H Test

This reflects statistical analysis of metabolomic data gathered from three different samples which are divided into three groups: Healthy, Pre-disease, and Disease. Each analysis is aimed to find a significant difference at the levels of the metabolites within these groups. The two tests utilized in this paper are:

ANOVA This test determines whether a significant difference exists among group means.

The **Kruskal-Wallis test** is a nonparametric alternative for use with ANOVA under violation of normality assumptions. To validate the ANOVA result, the Kruskal-Wallis H test, a non-parametric alternative, was used to compare the distributions of Metabolite_1 across the three groups.

Statistical analysis was conducted to evaluate the significance of the differences in levels of the metabolites between these groups for 10 metabolites, which were labeled as Metabolite_1 to Metabolite_10

Summary of Statistical Analysis:

	Metabolite	Test	p_value	Significance
1	Metabolite_1	ANOVA	0.637255907	Not Significant
2	Metabolite_1	Kruskal-Wallis	0.664703589	Not Significant
3	Metabolite_2	ANOVA	0.732163819	Not Significant
4	Metabolite_2	Kruskal-Wallis	0.834591682	Not Significant
5	Metabolite_3	ANOVA	0.553800823	Not Significant
6	Metabolite_3	Kruskal-Wallis	0.547535660	Not Significant
7	Metabolite_4	ANOVA	0.521987425	Not Significant
8	Metabolite_4	Kruskal-Wallis	0.515451272	Not Significant
9	Metabolite_5	ANOVA	0.965131640	Not Significant
10	Metabolite_5	Kruskal-Wallis	0.928474697	Not Significant
11	Metabolite_6	ANOVA	0.008084340	Significant
12	Metabolite_6	Kruskal-Wallis	0.009784189	Significant
13	Metabolite_7	ANOVA	0.646963681	Not Significant
14	Metabolite_7	Kruskal-Wallis	0.672189991	Not Significant
15	Metabolite_8	ANOVA	0.769963381	Not Significant
16	Metabolite_8	Kruskal-Wallis	0.714504237	Not Significant
17	Metabolite_9	ANOVA	0.901753149	Not Significant
18	Metabolite_9	Kruskal-Wallis	0.942238754	Not Significant
19	Metabolite_10	ANOVA	0.022316505	Significant
20	Metabolite_10	Kruskal-Wallis	0.026597828	Significant

Assumptions of ANOVA:

Independence: The samples from each of the groups of Healthy, Pre-disease, and Disease are independent of one another.

Normality: Levels of metabolites within every group are normally distributed.

Homogeneity of Variances: The variances of the levels of metabolites between the three groups should be roughly comparable.

Continuous data: The outcome of interest, such as the levels of metabolites, should be continuous.

Assumptions for the Kruskal-Wallis test are as follows:

1. Independence: The samples of three groups (Healthy, Pre-disease, Disease) are independent.
2. Data to Compare Needs to Be Ordinal or Continuous: It should be ordinal or continuous data for metabolite levels.
3. Same Distribution Shape: The levels of metabolites at each group must have the same kind of distribution. They may just vary in their medians.
4. No Normality Required: It does not require normal distribution of data whereas in ANOVA, data must be normally distributed.

Result Interpretation:

Metabolite 6:

p-value for ANOVA:0.0081 (significant)0.0098 Kruskal-Wallis p-value (significant).

Conclusion: Both tests point to a significant difference in the levels of Metabolite_6 between the three groups. This means that the illness condition has an effect on Metabolite_6.

Metabolite 10:

ANOVA p-value: 0.0223 (significant) p-value for Kruskal-Wallis: 0.0266 (significant)

Conclusion:

Metabolite_10 is a clearly separately varying metabolite, just like Metabolite_6. This metabolite might also be used as a biomarker to differentiate health and disease states.

Metabolites, no significant differences:

For the other metabolites, (Metabolite_1 to Metabolite_5, Metabolite_7 to Metabolite_9), $p > 0.05$ were obtained from ANOVA as well as Kruskal-Wallis test. This suggests there are no statistically significant differences between the groups for those metabolites and may not be associated with the disease condition.

Final Conclusion:

Metabolite_6 and Metabolite_10 showed a significant difference among the groups with statistical analyses, which are possibly involved in distinguishing between a healthy, pre-disease, and disease condition. Thus, they could be potential biomarkers for early diagnosis or monitoring of disease progression.

Part 4: Chi-Square Test for Independence

Testing if Gender is a statistically significant predictor of Disease Status (Healthy, Pre-disease, or Disease).

Method: Pearson's Chi-square test with Yates' continuity correction.

Assumptions of the Chi-Square Test for Independence:

1. Independence: Each observation (participant) should come from only one category so that all responses are independent.
2. Categorical Data: The variables to be categorical must be both gender and disease status.
3. Expected Frequency: The expected counts in the contingency table should usually be in excess of 5 to make recommendations based on the results. Sufficient Sample Size: One must ensure there is a sufficient sample size to confirm if the results are valid.

Summary of Outcomes:

Chi-square statistic (Chi-square) df p-value0.1013 1 0.7503 p-value (0.7503):The p-value is above the traditional threshold of significance of 0.05.It means that Gender has not been observed to be significantly related to Disease Status.Chi-square statistic (0.1013):

The low value of chi-square indicates that the observed frequencies in the contingency table are too close to the expected frequencies if there had not been any association.

This further supports the conclusion that Gender and Disease Status are independent of each other in this dataset.

Conclusion:

The results indicate that Gender has no effect on the probability of falling into any one of the three disease classes, namely, Healthy, Pre-disease, or Disease.

Thus, from data analyzed, disease status is independent of gender.

Part 5: Correlation Analysis

Correlating microbial species to metabolites. The data will be searched for the potential interactions or correlations that could illuminate microbial effects on metabolic processes.

Pearson Correlation:

Data were fetched from columns which corresponded with different microbial species (Species_1, Species_2, ., Species_20) and metabolites (Metabolite_1, Metabolite_2, ., Metabolite_10).

A correlation matrix was computed using the Pearson method, which determines the linear relationship between these species and their metabolites.

Visualization:

Correlation Plot: Developed using the ggcorrplot package to show the correlation coefficient values.

Heatmap:

Developed with the pheatmap package that depicts the clusters of metabolites and species by respective correlation values.

Results:

Correlation Matrix:

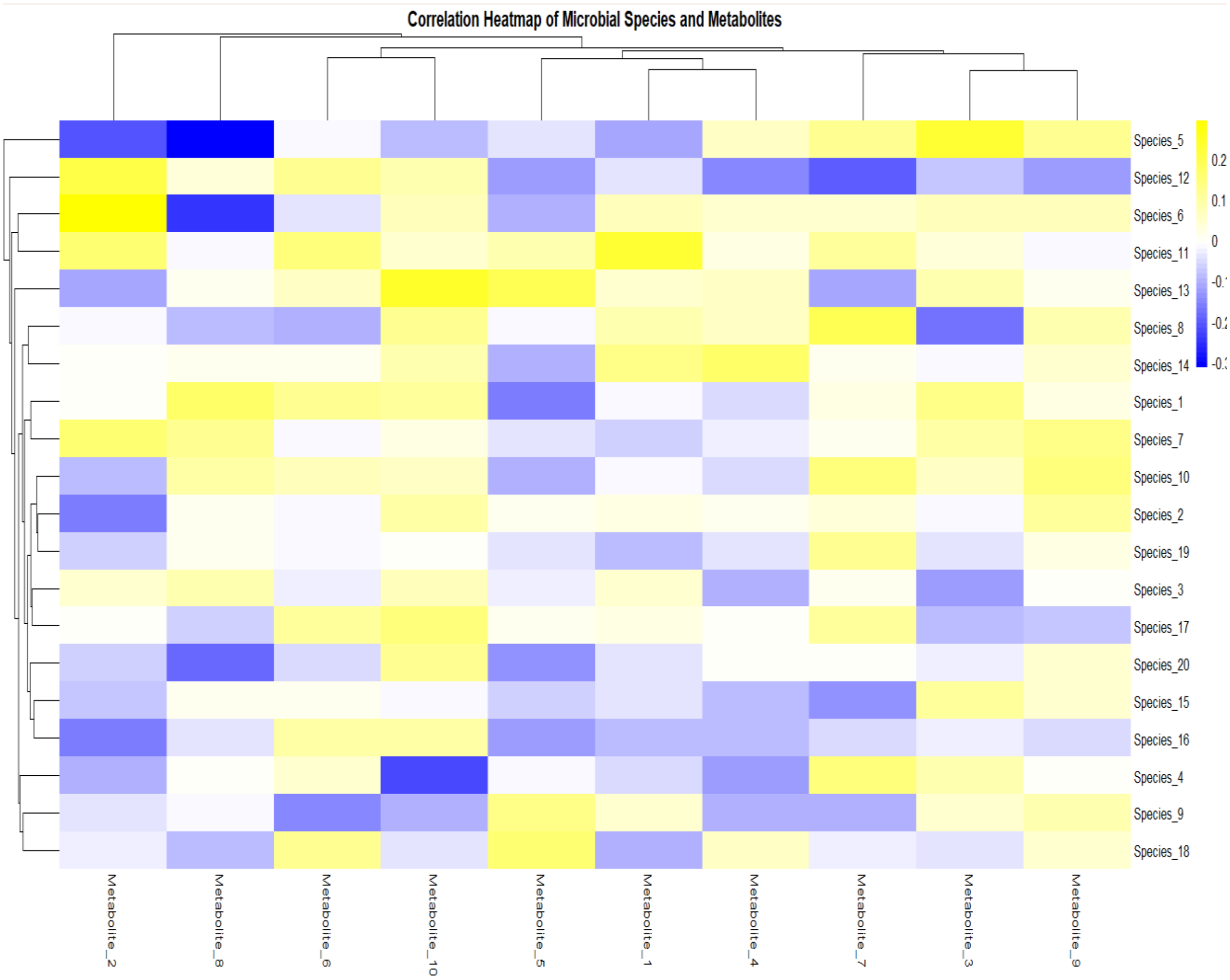
The table displayed the correlation coefficients, ranging from -1 to +1, between each microbial species and metabolite.Positive correlations were direct; i.e., increases in one variable went with increases in another. It approached +1.Negative correlations were inverse; i.e., increases in one variable had accompanying decreases in another. It approached -1.Weak correlations were close to 0 and suggested that there might not be a linear relationship.

Important Observations of the Correlation Matrix:

Most of the correlations were small, close to zero, suggesting a low abundance of strong linear relations for most species and metabolites.

Species_5 had some moderate values.With Metabolite_3 there was a positive correlation at $r = 0.223$ indicating one could be positively related Negative relationship with Metabolite_2 at r = -0.207, one could make an inverse association Species_6 had a moderate positive relationship at r = 0.293 with Metabolite_2. Species_11 revealed highly positive correlations with Metabolite_1, and also with Metabolite_2.

Visualization of the correlation plot:



The correlation plot, saved as correlation_plot.png contains the correlation coefficients with labels to facilitate interpretation. The heatmap illustrates the visual clustering of species and metabolites according to their correlation patterns:

Clusters of correlation profiles of species and metabolites with similar patterns suggest potential interaction networks. The blue-to-yellow gradient color scale represents the entire range of negative through to positive correlations.

Conclusion:

In the current dataset, it does not appear there is a strong linear association of microbial species with metabolites. The moderate correlations do exist, and these results are worthy of further exploration. Such observations indicate the complexity of such interactions between the microbiome and metabolome.

Part 6: Multivariate Analysis (MANOVA)

To identify interaction between the selected combination of microbial species and the metabolites that modulated specific dependent variables, namely BMI and Age. The general effect of these independent variables was compared by applying the MANOVA test.

Selection of Variables:

1. Established the first 5 from among the detected microbial species (Species_1, Species_2, ., Species_5).
2. Established the first 5 from among the metabolite variable candidates (Metabolite_1, Metabolite_2, ., Metabolite_5).
3. Generated the dataset by combining the chosen species and metabolites by forming independent_vars.
4. Established that the dependent variables (dependent_vars) are BMI and Age.

MANOVA Test:

Pillai's trace test statistic was utilised in conducting a **MANOVA** to identify if the combination of selected species and metabolites has statistically significant effects on the dependent variables. The Pillai's trace test statistic has been proven to be robust especially with unbalanced sample sizes or even when some of the assumptions were not met.

Results:

p-value: 0.5673

Assumptions for the test in MANOVA

1. Independence: Observations are independent of one another.

2. Multivariate Normality: The dependent variables, namely BMI and Age, should be normally distributed for every independent variable of microbial species and metabolites.
3. Homogeneity of Covariance Matrices: The variance-covariance matrices for the dependent variables should be homogeneous across groups.
4. No multicollinearity: The independent variables should not be significantly correlated with each other.

Interpretation of Results:

Pillai's trace value is 0.18665; independent variables, such as species and metabolites, account for about 18.67% of the variation in the dependent variables, BMI and Age.

But the p-value = 0.5673 is large by an order of magnitude beyond the standard level of significance, such as 0.05, and therefore the relationship between independent and dependent variables is statistically not significant.

The near F-value of 0.91609 with corresponding degrees of freedom also shows the lack of evidence for an effect.

Conclusion:

Results of MANOVA show that none of the chosen species of microbes and metabolites has an effect on BMI and Age, as a significant collective impact of them on these variables is not present in the dataset under consideration.

These results may indicate that other variables may be the more important correlate of BMI and Age, or that a different group of species and metabolites should be investigated to obatin this.
